higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 3

14 November 2019

This marking guideline consists of 9 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	B		
	1.1.2	C		
	1.1.3	B		
	1.1.4	D		
	1.1.5	A		
	1.1.6	A		
	1.1.7	B		
	1.1.8	C		
	1.1.9	D		
	1.1.10	D		
			(10 × 1)	(10)
1.2	1.2.1	D		
	1.2.2	L		
	1.2.3	F		
	1.2.4	C		
	1.2.5	I		
	1.2.6	O		
	1.2.7	H		
	1.2.8	A		
	1.2.9	M		
	1.2.10	E		
			(10 × 1)	(10)
1.3	1.3.1	Request for proposal		
	1.3.2	Management information system		
	1.3.3	Request for quotation		
	1.3.4	Chief operating officer		
	1.3.5	Chief executive officer		
	1.3.6	Transaction processing system		
	1.3.7	Office automation system		
	1.3.8	Point of sale		
	1.3.9	Database management system		
	1.3.10	Executive information system		
			(10 × 1)	(10)

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1.4 1.4.1 Ergonomics
1.4.2 High-level
1.4.3 Resource management
1.4.4 Attributes
1.4.5 Data store
1.4.6 Lease
1.4.7 Financing
1.4.8 Strategic
1.4.9 System
1.4.10 Analysis

(10 × 1) (10)
[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

2.1 An information system (IS) is an organised system for the collection, organisation, storage and communication of information.

OR

An information system (IS) is a system in which the goal or purpose is to produce information, which can be used by the business. (2)

2.2
- Stage
- Steps
- Techniques and skills
- Outcomes and deliverables
- Quality assurance
- Cyclical approach
- Well-formed structures (Any 4 × 1) (4)

2.3
- Lack of user involvement
- Technological problems
- Lack of expertise
- Lack of understanding of the problem, opportunity or directive
- Incomplete specification of requirements and functionality
- Changing requirements
- Lack of management support for the project
- Lack of control or planning (8)

2.4 Phased development approach
The overall system is broken down into a series of versions, each of which is developed sequentially.

Waterfall approach
The development team moves sequentially through the various stages of the SDLC, completing each stage before moving on to the next one. (2 × 2) (4)
[18]

QUESTION 3

3.1 Data processing is the collection and manipulation of items of data to produce meaningful information. (2)

3.2 A transaction processing system (TPS) is an information processing system✓ for business transactions,✓ involving the collection, modification and retrieval of all transaction data.✓ (3)

3.3
- Organising
- Analysing
- Manipulating (3)

3.4
- Procedural reports
- Exception reports
- Scheduled reports
- Ad-hoc reports (Any 2 × 1) (2)

3.5 Operational systems✓
Systems that support the operational side of a business✓

Managerial systems✓
Systems that support the planning and decision-making side of a business✓

Strategic systems✓
Systems that provide support for the competitive advantage of an organisation by enabling the use of competitive strategies✓ (6)
[16]

QUESTION 4

4.1 4.1.1 A chief knowledge officer is an organisational leader who is responsible for ensuring that the organisation maximises the value it achieves through knowledge.

4.1.2 A webmaster is the person who is responsible for maintaining one or many websites.

4.1.3 Chief information officer (CIO) is an executive job title commonly given to the person who is in charge of the information technology (IT) strategy and the computer systems required to support the organisation's unique objectives and goal.

4.1.4 The chief technology officer (CTO), sometimes known as the chief technical officer or chief technologist, is an executive-level position in a company or other entity that is concerned with scientific and technological issues in an organisation.

4.1.5 A systems analyst is a person who analyses a complex process or operation to improve its efficiency, especially by applying a computer system.

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4.1.6 A database administrator is responsible for the performance, integrity and security of a database.

4.1.7 A computer programmer is the person who writes computer programs.

4.1.8 A network administrator is the person who is responsible for maintaining computer infrastructures, with the emphasis on networking.

4.1.9 A mobile app developer is a professional who is focused on developing software or systems for mobile devices.

(9 × 2) (18)

4.2
- Data capture and entry
- Production control and support
- Hardware and operations
- Performance monitoring system tuning and capacity planning
- Job scheduling
- System back-ups
- System audits
- Disaster recovery and business continuity planning (Any 5 × 1) (5)

4.3 Operations support systems
Complex technical problems require expertise in specific areas or technologies.✓ These are dealt with by specialists.✓

Systems support
Technical experts support application analysis and design for new information systems.✓ The development team would not be expected to have expertise in all areas of technology, and they would therefore call on experts in databases, networking, specialists in hardware or software, as required.✓ (4)

[27]

QUESTION 5

5.1 Rental✓
It gives an organisation full use of the computer technology for an agreed period of time at an agreed monthly cost.✓

Leasing✓
It is similar to a rental, but it normally lasts for a longer period and it is used for high-value assets.✓

Purchasing✓
It enables an organisation to borrow the capital needed to buy the computer technology and pay the loan back over a period of time.✓

(Accept any relevant explanation) (6)

5.2
- Specify the information systems requirement.
- Create and issue a request for proposal or a request for quotation.
- Conduct a preliminary evaluation of the proposals.
- Conduct a detailed evaluation of the short-listed proposals.
- Award the contract on the overall proposal.
- Implement the information systems technology. (6)

5.3 5.3.1 Heat can cause a computer to run more slowly, damage components, and in the case of laptops, drastically shorten battery life.

5.3.2 Dust build-up can block the air vents and create an insulating layer on top of heat-generating parts, putting them at risk of overheating.

5.3.3 Magnets can damage the monitor and hard drive of the computer.

5.3.4 Power surges can burn the motherboard.

5.3.5 The internal parts of a computer are susceptible to static electricity, especially the hard drive, and it can cause considerable damage to the hard drive if it is zapped by even a small amount of electrostatics.

(5 × 2) (10)

5.4 Information systems outsourcing
The contracting or subcontracting of non-core activities to free up cash, personnel, time, and facilities for activities in which a company has a competitive advantage

Information systems off-shoring
The moving of various operations of a company to another country for reasons such as lower labour costs or more favourable economic conditions in the other country

(2 × 2) (4)
[26]

QUESTION 6

6.1 An organisation is a group of people working together to achieve a common goal. (2)

6.2 6.2.1
- Top level management take strategic decisions.
- They are long-term decisions.
- For example, the top managers of a bakery need to take a strategic decision about whether to remain in the cafe business.

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6.2.2
- Middle level management take tactical decisions
- They help implement the strategy of top management.
- For example, a simple tactical decision for the cafe would be whether to open earlier in the morning or on Saturdays to attract new customers.

6.2.3
- Lower level management take operational decisions.
- They see to the day-to-day running of the business.
- For example, a simple operational decision for the cafe would be whether to order more coffee for the following week.

(3 × 3) (9)

6.3 Positional analysis is the means by which the position of an organisation is examined regarding its state. (3)

6.4
- Market position
- Financial position
- Sales position
- Operations position
- Distributions position
- Research and development position
- Production position
- Technology or information systems position (Any 5 × 1) (5)

6.5
- Cost leadership strategy
- Differentiation strategy
- Innovation strategy
- Growth strategy
- Alliance strategy (5)

6.6 Step 1: Identify and understand the business drivers.
Step 2: Create the information systems vision.
Step 3: Define the gap (strategic information systems planning).
Step 4: Define the planning approach.
Step 5: Describe the business need of the organisation.
Step 6: Define the data requirements of the organisation.
Step 7: Create the information technology blueprint.
Step 8: Do information technology tactical planning. (8)

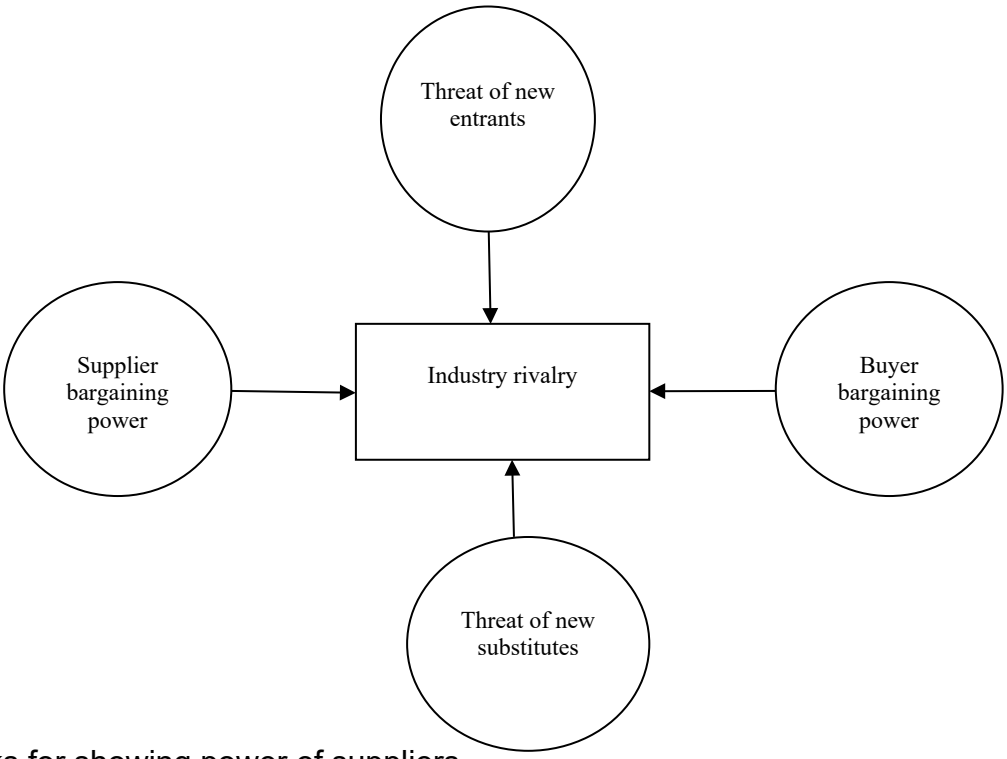
6.7 Cash cow✓
A product in a low-growth area which has a high market share✓

Star✓
A product in a high-growth market in which the company has a large market share✓

Dog✓
A product in a low-growth market for which the company has a low market share✓

Problem child✓
A product in a high-growth market that has a low market share✓ (4 × 2) (8)

6.8



2 marks for showing power of suppliers
2 marks for showing power of buyers
2 marks for showing industry rivalry/competitors
2 marks for showing threat of substitutes
2 marks for showing threat of new entrants

(Any other relevant answer) (10)
[50]

QUESTION 7

7.1 7.1.1 A process by which business decisions are analysed

7.1.2 The total value of the benefits minus the total value of the costs, with both costs and benefits adjusted to reflect the point in time at which they occur

7.1.3 A document that describes the company, lists the IT services or products needed, and specifies the features required

(3 × 2) (6)

7.2 ROI = (Total benefits – Total costs) / Total costs

OR

The basic formula for ROI is: ROI = Net Profit / Total Investment * 100 (2)

7.3 7.3.1 R520 000
7.3.2 R1 100 000
7.3.3 R160 000
7.3.4 R400 000
7.3.5 –R800 000

(5 × 1) (5)

7.4
- Identify all the potential costs and benefits.
- Allocate values or estimates to the costs and benefits.
- Examine the cash flow of the project.
- Calculate the net present value of the project.
- Calculate the return on investment of the project.
- Determine the break-even point of the project. (6)

7.5 Project scoping
It involves creating boundaries that define what is included and what is not included in the project. It defines exactly what is needed in order to accomplish the project.

Problem identification
It is a process of problematising the current situation in order to understand what the current state is or what problems are occurring in the current system within the organisation. (2 × 2) (4)

[23]

TOTAL SECTION B: 160
GRAND TOTAL: 200